IN A HIGH-DIMENSIONAL PRODUCT GRAPH, TWO ISOLATED VERTICES OF THE RANDOM SUBGRAPH ARE TYPICALLY AT DISTANCE AT LEAST THREE

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Editorial note. The constant alpha_M and the parameters n, d and p that the proof uses are set up in the author's model description earlier in the paper; the editorial prerequisite list on the first page names them and gives the source location, and no author text was moved into the counted spans to replace them.

Allowed standard prerequisites (editorial).

- The parameters of the model: n the order and d the dimension-regularity parameter of the product graph, p the edge probability, and the constant alpha_M controlling the one-edge overlap factor. Used inside the counted proof at source lines 494-508; the model and the relation d = Theta_M(log n) are set up earlier in the paper (around source lines 201-240). Gives the two counts of ordered pairs (nd adjacent pairs and nd squared pairs at distance two) and the constant alpha_M in the bound (1-p)^(2d-1) <= alpha_M (1-p)^(2d).

Lemma 0.1. *Whp, every two isolated vertices of G_p are at distance at least three in G.*

$$d^{-1/50} \le n(1-p)^d \le d^{1/50}. \tag{1}$$

Proof. There are at most nd ordered adjacent pairs and at most nd^2 ordered pairs at distance two. If distinct vertices u, v are not adjacent, the sets of edges incident with them are disjoint, and hence

$$\mathbb{P}(u, v \text{ are isolated in } G_p) = (1-p)^{2d}.$$

If u and v are adjacent, their incident edge sets overlap in one edge, so this probability is

$$(1-p)^{2d-1} \le \alpha_M (1-p)^{2d},$$

for some constant α_M. The expected number of ordered pairs of isolated vertices at distance at most two is therefore at most

$$\alpha_M n d^2 (1-p)^{2d} \le \frac{\alpha_M d^{2+1/25}}{n} = o(1),$$

where we used that p satisfies (1). We conclude with Markov's inequality. □

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